Xuqin **Wang**

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Research Interests

Generative AI, representation learning, and multimodal learning, with broader experience in domain adaptation, image editing, generative-model distillation, 3D vision, scene understanding, and agentic AI.

Education

Technical University of Munich — *Munich, Germany*

Ph.D. student, School of Computation, Information and Technology — *May 2023 – Present*

- Supervisors: Prof. Dr. Daniel Cremers and Prof. Dr.-Ing. Niclas Zeller.

Télécom Paris, Institut Polytechnique de Paris — *Palaiseau, France*

Diplôme d'ingénieur (M.S.), Image Processing and Data Science — *Sep. 2020 – Nov. 2022*

Shanghai Jiao Tong University — *Shanghai, China*

B.S., Computer Science and Technology — *Sep. 2016 – Jul. 2020*

Publications

Adaptively Pivoted Flow Matching for Compositionally Consistent Translation

Wang, X., Chen, X., Zeller, N., Cremers, D., & Wu, T. — *Under Review*

Controlled image translation requires preserving source appearance and consistency across sequential condition changes. We formulate translation through a shared latent pivot as a translation groupoid and introduce Adaptively Pivoted Flow Matching (AP-FM), which learns adaptive Gaussian priors to reduce conditional velocity ambiguity. The approach improves target fidelity, source appearance preservation, and compositional consistency on novel-view synthesis and pose-guided appearance transfer.

VideoReloc: Long-Term Indoor Video Relocalization against a Kilobyte-Scale Semantic Scene Graph

Li, Q., Chen, X., **Wang, X.**, Zhang, Z., Luo, H., Wu, T., Cremers, D., Liu, L., & Zhang, Y. — *Under Review · arXiv:2609.21804*

Long-term indoor visual relocalization is challenged by changes in lighting and furniture layouts. VideoReloc uses compact semantic scene graphs (approximately 100 kB per room) instead of appearance-dependent maps, combining adaptive video clips, hypothesis-first registration, and orientation-aware refinement. It improves localization robustness and accuracy on RIO10 and ReplicaCAD while reducing map storage by roughly two orders of magnitude.

GeodesicNVS: Probability Density Geodesic Flow Matching for Novel View Synthesis

Wang, X., Wu, T., Zhang, Y., Liu, L., Sun, M., Wang, Y., Zeller, N., & Cremers, D. — *CVPR 2026 · arXiv:2603.01010*

We introduce probability-density-aware geodesic flow matching to model view transitions along density-consistent trajectories. The approach improves geometric consistency and view-synthesis quality while providing a principled connection between probability geometry and generative learning.

CinemaTraj: Composing Atomic Camera Trajectories for 3D Scenes with LLM Agents

Li, Q., Chen, X., Türköz, E., Liu, L., **Wang, X.**, Meng, L., Wu, T., & Zhang, Y. — *ACM MM 2026 · arXiv:2607.26910*

CinemaTraj equips an LLM agent with a structured 3D scene graph and decomposes user instructions into atomic camera movements, generating collision-free, prompt-aligned camera trajectories with synchronized narrative content.

MeSS: City Mesh-Guided Outdoor Scene Generation with Cross-View Consistent Diffusion

Chen, X., Zhai, Z., Zhou, K., Wang, Z., He, J., Wang, D., Zhang, Y., Sun, M., **Wang, X.**, Westermann, R., Wu, T., Schindler, K., & Meng, L. — *DAGM GCPR 2026 · arXiv:2508.15169*

MeSS generates a cross-view-consistent 3D Gaussian Splatting scene from mesh-rendered geometric and semantic priors, improving visual consistency and geometry alignment on a city-scale benchmark and transferring zero-shot to a real LoD3 city mesh.

GeoSceneGraph: Geometric Scene Graph Diffusion Model for Text-Guided 3D Indoor Scene Synthesis

Ruiz, A., Wu, T., Melnik, A., Cheng, Q., **Wang, X.**, Liu, L., Wang, Y., Zhang, Y., & Ritter, H. — *ISPRS 2026 (Oral) · arXiv:2511.14884*

GeoSceneGraph combines structured scene-graph representations with diffusion-based generation to explicitly encode semantic and geometric relationships between objects, improving controllability and realism for indoor scene synthesis.

LADB: Latent Aligned Diffusion Bridges for Semi-Supervised Domain Translation

DAGM GCPR 2025 (Oral) ·
arXiv:2509.08628

Wang, X., Wu, T., Zhang, Y., Liu, L., Wang, D., Sun, M., Wang, Y., Zeller, N., & Cremers, D.

We propose latent aligned diffusion bridges to learn transport between source and target latent spaces while leveraging abundant unpaired target data, enabling more effective cross-domain generation under weak supervision.

Experience

Huawei Munich Research Center *Munich, Germany*

Industrial PhD Researcher *May 2023 – Present*

- Conduct industrial Ph.D. research in generative AI and computer vision, spanning problem formulation, model development, large-scale experimentation, scientific publication, and collaboration with academic and industrial research teams.
- Research experience includes diffusion and flow-based generative models, image translation and editing, novel-view synthesis, 3D scene understanding, multimodal learning, model distillation, and agentic AI.
- Build and maintain distributed training, evaluation, and benchmarking pipelines for reproducible model development and experimentation.

Tremau *Paris, France*

Research Intern · Advisor: Theodoros Evgeniou *Mar. 2022 – Apr. 2023*

- Conducted applied computer-vision research for online content safety, with experience in dataset curation, visual retrieval, image-similarity modeling, pipeline development, and multi-dataset evaluation.

Skills

ML systems	Python, C++, PyTorch, CUDA, Diffusers, LoRA, Transformers, DeepSpeed, distributed training (DDP/FSDP), Slurm, Docker, Git, and Weights & Biases.
Generative models	Image/video diffusion, flow matching, diffusion bridges, and diffusion distillation
3D vision	Novel view synthesis, 3D reconstruction, 3D Gaussian Splatting, Neural Radiance Fields, multi-view geometry, scene understanding, and open-vocabulary 3D
Agentic AI	LiteLLM, smolagents, vLLM
Languages	Chinese (native), English (fluent), French (intermediate), German (beginner)

Community Contributions

2024 **Organizer,** ECCV 2024 Workshop on Spatial AI